ADHAVAN PORNAM

Dindigul, Tamil Nadu | +91 8825473617 | Gmail | LinkedIn | GitHub | Portfolio

A driven Data Science and Machine Learning enthusiast with a strong foundation in analytics, data management, and SQL. Hands-on experience building predictive models using Python, with expertise in deep learning (CNN, RNN), NLP, and data visualisation. Demonstrated teamwork and leadership as a student Community Builder. Passionate about leveraging data to build technical solutions and make a positive impact on society.

EDUCATION

Imarticus Learning, Online *2025 – 2026*
Post Graduate Program in Data Analytics

SRM Institute of Science and Technology, Chennai *Sep 2021 – Dec 2025*
Bachelor of Technology, Computer Science and Engineering *(CGPA 7.3)*

TECHNICAL SKILLS

Programming: Python, SQL
Machine Learning: Scikit-learn, Regression, Classification, Clustering, Random Forest, XGBoost, SVM (SVC), Cross-Validation, ROC-AUC, Confusion Matrix, Hyperparameter Tuning
Deep Learning: TensorFlow, Keras, CNN, BiLSTM, RNN/LSTM, Conv1D, ReduceLROnPlateau
Time Series: LSTM, SARIMA, SARIMAX, Auto ARIMA (pmdarima), MinMaxScaler, Time-Step Sequencing, RMSE, yfinance
NLP: NLTK, Text Preprocessing, Tokenization, Label Encoding
Data Analysis: Pandas, NumPy, Matplotlib, Seaborn, EDA, PCA, StandardScaler, Dimensionality Reduction, Statistical Analysis, Data Cleaning
BI & Visualization: Power BI, DAX, Data Modelling, Tableau, Canva, Data Bricks
Deployment & Tools: Flask, Gunicorn, OpenCV, Git, GitHub, LLM APIs (Claude, GPT-4o, Gemini), Ollama (local LLMs), Langflow, RAG (Retrieval-Augmented Generation), Prompt Engineering
Cloud Platforms: AWS EC2 (Ubuntu, T3), Gunicorn, tmux, Vercel, REST API Integration, PySpark (local), Databricks (Basics)

PROJECTS

Consumer Complaint Classifier | Deep Learning & NLP *2026*

- Developed an end-to-end NLP classification system on **~45,000** balanced CFPB complaint samples across **9 financial product categories**; benchmarked **4 architectures** (SimpleRNN, LSTM, BiLSTM, **CNN-BiLSTM**) — the hybrid **CNN-BiLSTM** achieved **80.43% test accuracy** and **0.81 macro F1-score**.
- Engineered the model with **SpatialDropout1D** after embedding, **Conv1D** for local n-gram pattern extraction, and a **Bidirectional LSTM** for global sequence context; added a **low-confidence guard (<20%)** in the Flask app to prevent unreliable predictions from being shown to users.
- Deployed the Flask application on **AWS EC2 (Ubuntu, T3)** with **Gunicorn (2 workers)** and **tmux** for persistent process management; integrated **Xano REST API** for real-time complaint logging, confidence scores, and **unique Complaint ID** generation to a cloud database.
- Tools: Python, TensorFlow/Keras, NLTK, Pandas, Scikit-learn, Flask, AWS EC2, Gunicorn, tmux, Xano REST API, WinSCP, PuTTY.

Facial Emotion Recognition System | CNN & Computer Vision *2026*

- Trained a Sequential CNN on the **FER-2013** dataset (**~35,887 grayscale images** resized to **96×96px**, **7 emotion classes**) using **3 Conv2D layers** for spatial feature extraction, Dropout (20%) for regularisation, and **ReduceLROnPlateau** to improve convergence — achieving **90% overall accuracy**.
- Integrated OpenCV's **Haar Cascade Classifier** (**Viola-Jones algorithm**) for real-time face localisation on live camera feed; the detected region is cropped and passed to the CNN for per-frame emotion classification — achieving strong per-class F1-scores (**Happy: 0.94**, **Surprise: 0.92**).
- Tools: Python, TensorFlow, Keras, OpenCV, Haar Cascade, Pandas, Matplotlib, Kaggle (FER-2013).

Stock Market Prediction using RNN-LSTM | Neural Networks *2026*

- Built a **3-layer Stacked LSTM** (50 units each, **Adam optimiser, MSE loss**) on **5 years** of Titan Company (NSE) daily closing price data; applied **MinMaxScaler** normalisation and structured input sequences using a **100-day time-step window** on a **75/25 sequential train-test split** (**926 training / 309 test samples**).

- Evaluated model performance using **RMSE (train: 2,881 / test: 3,701)**; generated a **20-day iterative forward forecast** with inverse-transformed price predictions and visualised historical vs. predicted trends with **confidence bands** using Matplotlib.
- Tools: Python, TensorFlow/Keras, Scikit-learn (MinMaxScaler, RMSE), NumPy, Matplotlib, yfinance.

Investment Intelligence Dashboard | Power BI *2026*

- Integrated **4 open government datasets** — MSME UDYAM registrations (**788 districts, 36 states**), RBI credit data, NPCI UPI transactions (**1,258 rows**, FY 2023–26), and NFS district-level volumes (**11,280 rows**) — into a **star-schema data model** using LGD district code as the common geographic key.
- Engineered a **weighted Investment Readiness Score** in DAX — **(MSME Score × 0.40) + (Credit Score × 0.40) + (Digital Score × 0.20)**, with **min-max normalisation** — classifying all districts into **4 investment tiers**; built a **5-page interactive dashboard** to enable banks, policymakers, and investors to identify **credit cold spots** and high-potential MSME zones.
- Tools: Power BI Desktop, Power Query (M), DAX, Star Schema Modelling, UDYAM / RBI / NPCI Open Government Data.

TN PHC Medicine Stockout Predictor | PySpark MLlib & ALS *2025*

- Identified a high-value public health problem — medicine stockouts at Tamil Nadu Primary Health Centres (PHCs) — and engineered a proxy Utilization Score (Stock Distributed ÷ Total Stock) as a stockout risk label; classified districts as HIGH RISK if utilization ≥ 0.85 or stock = 0.
- Built a PySpark MLlib ALS (Alternating Least Squares) collaborative filtering pipeline on 5 years of HMIS open government data (2013–2020) across Tamil Nadu districts and key essential medicines; structured as a District × Medicine × Utilization matrix to predict per-district stockout risk one month in advance.
- Evaluated pipeline using RMSE and MAE; generated ranked district-medicine risk recommendations enabling health officers to trigger proactive reorders before patients are turned away.
- Tools: PySpark (MLlib, SQL, Pipeline API), ALS Collaborative Filtering, Java 17, HMIS Open Government Data, Pandas, Matplotlib.

ACHIEVEMENTS & VOLUNTEERING

Coffee Inc, Chennai *Feb 2023 – May 2023*

Intern, Business & Web Development

- Developed and **deployed Flask web applications to automate workflows**, reducing manual data entry and improving operational efficiency.

Teedian Student Community, Chennai *Jun 2023 – Aug 2024*

Marketing Intern & Community Manager

- Designed and executed **data-driven marketing campaigns**, **tracking engagement metrics** to **measure campaign performance** and optimise content strategy.
- Created and managed social media content calendars; **analysed audience insights and post analytics** to improve reach and community engagement.
- Produced promotional material for community events; **documented campaign outcomes and reported performance trends** to the leadership team.

Think Digital Student Club, SRM Institute of Science and Technology *Dec 2022 – Aug 2024*

Machine Learning Domain Member

- Collaborated with a cross-functional team to **build and deploy NLP, data analysis**, **and machine learning projects**; deployed solutions on **Vercel as part of the club's applied AI initiatives**.

CERTIFICATIONS & LANGUAGES

Data Strategy — 365 Data Science *Nov 2022*

Data Literacy — 365 Data Science *Sep 2022*

Broadridge India Innovation Hackathon Aug 2023

Languages: English (Professional), Tamil (Native)

PERSONAL INFORMATION:

Address: **Ramar Colony, NGO Colony, Dindigul 624004**

Date of Birth: 26/03/2002